

3H Reconstruct a String from its k -mer Composition

String Reconstruction Problem

Reconstruct a string from its k -mer composition.

Input: A collection of k -mers *Patterns*.

Output: A string *Text* with k -mer composition equal to *Patterns*.

AT CA GG GG GT TG TG
↓
CATGGGTG

Formatting

Input: A space-separated list of k -mer strings *Patterns*.

Output: A string *Text* with k -mer composition equal to *Patterns* (if multiple answers exist, you may return any one).

Constraints

- The number of strings in *Patterns* will be between 1 and 10^4 .
- The length of any one pattern in *Pattern* will be between 1 and 10^2 .

Test Cases

Case 1

Description: The sample dataset is not actually run on your code.

Input:

3

ACG CGT GTG TGT GTA TAT ATA

Output:

ACGTGTATA

Case 2

Description: The sample dataset is not actually run on your code.

Input:

2

GG AC GA CT

Output:

GGACT

Case 3

Description: The sample dataset is not actually run on your code.

Input:

3

AAC AAC ACG ACT CGA GAA

Output:

AACGAACT

Case 4

Description: The sample dataset is not actually run on your code.

Input:

4

CTAC CTCC TCCT ACTC CCTC CCTA TACT

Output:

CCTACTCCTC

Case 5

Description: The sample dataset is not actually run on your code.

Input:

3
CCC CCC CCC TCC CCC CCG CCC CCC CCC

Output:

TCCCCCCCCCG

Case 6

Description: The sample dataset is not actually run on your code.

Input:

2
AG AT AA GA GG GT TA TG TT AT

Output:

AAGTTGGATAT

Case 7

Description: The sample dataset is not actually run on your code.

Input:

3
ACG CGT GTA TAC

Output:

ACGTAC

Case 8

Description: A larger dataset of the same size as that provided by the randomized autograder.